Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
C		2619940

Fan Voting Estimation in DWTS based on Dynamic Bayesian Hierarchical Model

"Dancing with the Stars" (DWTS) combines professional judges' scores with audience voting to determine weekly eliminations of contestants. However, fan vote counts are unobservable, making it difficult to distinguish genuine popularity from statistical noise. This study develops an integrated framework to reconstruct hidden fan vote, evaluate existing voting rules, analyze key influencing factors, and design an improved adaptive voting scheme.

Task 1: Fan Vote Estimation. We formulate the inference of weekly fan vote shares as a Bayesian inverse problem. A **Dynamic Bayesian Hierarchical Model** is constructed, the elimination outcomes are incorporated into the likelihood, and posterior inference is performed using **MCMC**. The reconstructed fan votes achieve high agreement with observed eliminations, with **consistency of 87.0%** under Percent rule and **79.7%** under Rank rule. Uncertainty, measured by the **relative 90% credible interval width (RCI90)**, contracts over the season from **47%** in early weeks to about **30%** near the finale.

Task 2: Voting Scheme Comparison. Using the inferred fan vote shares, we conduct counterfactual reconstructions under **Rank-based** and **Percent-based** aggregation rules. While the two schemes exhibit strong agreement (Kendall $\boldsymbol{\tau} = \mathbf{0.692}$ and Spearman $\boldsymbol{\rho} = \mathbf{0.783}$), the Percent scheme shows stronger alignment with fan preferences (alignment **0.740** v.s. **0.328** for Rank). Meanwhile, the Rank scheme is more stable under fan voting (sensitivity **0.112** v.s. **0.218** for Percent). An **AHP** analysis over four features yields a consistent judgment under which the **Percent** method ranks first (score **0.8399**).

Task 3: Impact Analysis of Celebrity Characteristics. To quantify the influence of celebrity attributes and professional partners, we build separate **Random Forest** regression models for fan voting and judges' scoring. With season-grouped cross-validation and lagged features using only historical information, the models achieve $\boldsymbol{R^2} \approx \mathbf{0.80}$ for fans and $\boldsymbol{R^2} \approx \mathbf{0.90}$ for judges. Feature importance and **SHAP** analyses indicate that structural competition variables dominate predictive power, while ballroom partners, celebrity age, and industry background exert consistent secondary effects. Partner effects exhibit strong heterogeneity and are more pronounced in the fan voting model.

Task 4: Improved Voting System Design. To address the trade-off between fairness and entertainment, we propose the **Dynamic Weighting - Uncertainty Weighted (DWUW)** system. This adaptive mechanism adjusts the weight of judges' scores based on the uncertainty of fan voting signals. A conditional judges-save rule is also introduced to prevent large technical mismatches. We show that DWUW achieves high fairness, with season-level correlation $|r_s| \approx$ **0.98**, while maintaining strong engagement through sustained upset rates of approximately **68%**.

Conclusion. By integrating Bayesian inference, comparative rule evaluation, machine learning and adaptive mechanism design, this study provides a comprehensive solution to the DWTS voting problem. The proposed **DWUW** framework is broadly applicable to decision systems combining expert judgment with crowd-based input.

Keywords: Dynamic Bayesian Hierarchical Model, MCMC, AHP, Random Forest, SHAP, DWUW.

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1 Introduction

1.1 Problem Background

"Dancing with the Stars" (DWTS) is a globally renowned franchise that juxtaposes technical dance proficiency with celebrity popularity. The competition's core mechanic involves a dual-voting system: professional judges score contestants based on technical merit (typically on a scale of 1-10 per judge), while the viewing public votes based on preference via phone, text, or online platforms. Ideally, this hybrid system balances skill and entertainment value, creating a spectacle that rewards growth and charisma.

However, this balance has occasionally sparked controversy. Notable instances, such as NFL legend Jerry Rice reaching the finals despite consistently low judge scores (Season 2) and radio personality Bobby Bones winning with consistently poor technical marks (Season 27), have highlighted the potential disconnect between professional evaluation and public sentiment. These events have prompted the show's producers to oscillate between different aggregation methods—specifically, **Rank-based** and **Percent-based** systems—and introduce **"Judge-save"** mechanisms.The central challenge lies in the opacity of the fan vote data, which is a closely guarded secret, making it difficult to objectively assess the fairness and consistency of these voting schemes.

1.2 Clarification and Restatement

The objective of this study is to unravel the dynamics of the DWTS voting system through data-driven modeling. Based on the problem description, we have structured our investigation into four main tasks:

- **Task 1: Fan Vote Estimation.** Develop a mathematical model to estimate the hidden fan votes and quantify the consistency and uncertainty of these estimates.
- **Task 2: Voting Schemes Comparison.** Compare the Rank v.s. Percent voting schemes, analyze specific controversial cases (e.g., Jerry Rice, Bobby Bones), and evaluate the impact of a potential "Judge-save" mechanism.
- **Task 3: Impact Analysis.** Evaluate the influence of ballroom partners and celebrity demographics (Age,Industry,etc) on voting outcomes.
- **Task 4: New System Proposal.** Design a novel, fairer, and more exciting voting system (the DWUW Model).

1.3 Overview of Our Approach

To address these challenges, we adopt a comprehensive multi-stage modeling framework, combining Bayesian inference, comparison methods, machine learning and the DWUW Model. Our workflow is visually summarized in Figure 1.

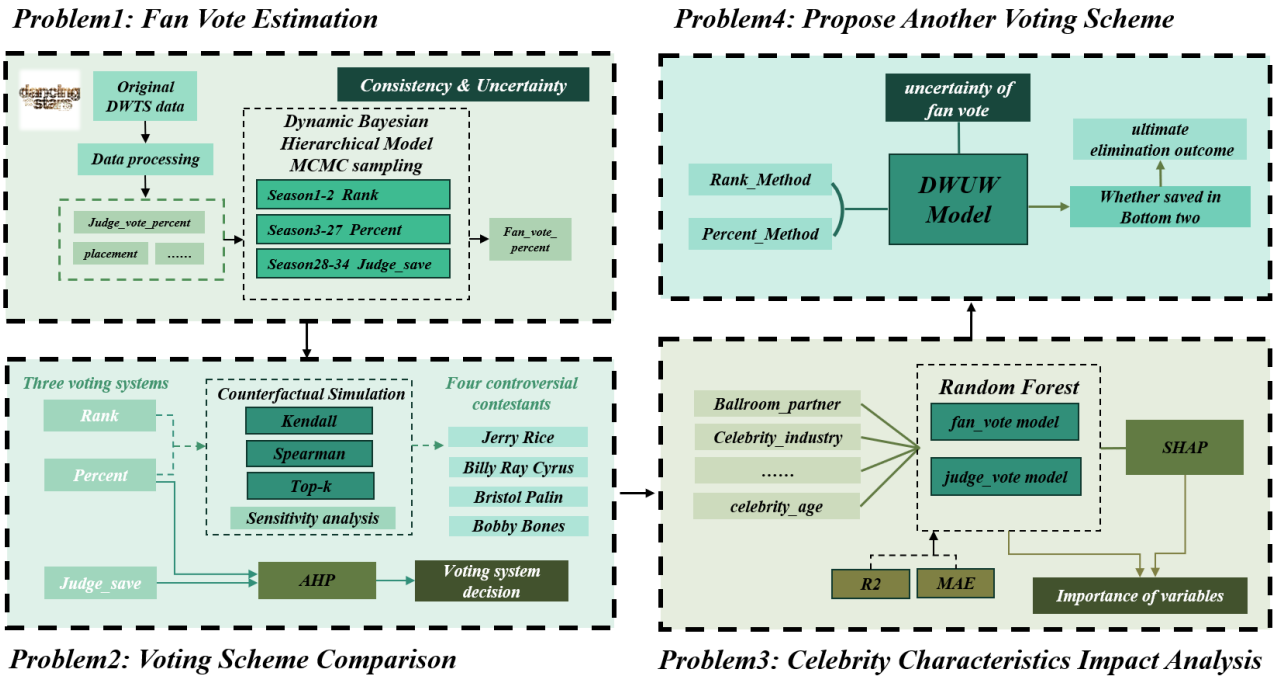


Figure 1: Our Work

Our approach follows these strategic steps:

- **Data Preprocessing:** We implement a rigorous cleaning pipeline to handle missing values, normalize heterogeneous scoring scales, and extract demographic features.
- **Bayesian Inference (Task 1):** We develop a **Dynamic Bayesian Hierarchical Model** to infer the posterior distribution of latent fan vote shares, where elimination outcomes are incorporated through **probabilistic likelihood-based constraints**. Posterior inference is carried out via MCMC sampling.
- **Comparative Analysis(Task 2):** We evaluate voting methods using multiple correlation indexes and the **Analytic Hierarchy Process (AHP)**, scoring them on stability, fairness, and transparency metrics derived from counterfactual simulations.
- **Machine Learning for Impact Analysis (Task 3):** We employ a **Random Forest** regression model interpreted via **SHAP (SHapley Additive exPlanations)** values to disentangle the non-linear impact of dancers and celebrity characteristics.
- **Dynamic Mechanism Design (Task 4):** We propose the **DWUW (Dynamic Weighting - Uncertainty Weighted)** system, which adaptively shifts power between judges and fans based on real-time signal quality.

2 Assumptions and Notation

2.1 Modeling Assumptions

To ensure the tractability of our models, we make the following assumptions:

- **Assumption 1: Rationality of Eliminations.** We assume the observed elimination results are the ground truth. Any estimated fan votes must mathematically result in the actual eliminated contestant having the lowest combined score (Rank or Percent).
- **Assumption 2: Stability of Fan Base.** A celebrity's fan base is assumed to be relatively stable week-over-week, modeled as a random walk process. Sudden, inexplicable shifts are penalized by the model prior.
- **Assumption 3: Independence of Judge Scores.** We assume judges score each dance independently based on its merit, without collusion.
- **Assumption 4: Consistent Rule Application.** For seasons where the specific voting method is ambiguous (e.g., transition periods around Season 28), we classify them based on the prevailing trend described in the problem statement (Rank-based).

2.2 Notation

We define the following notation:

Symbol	Description
s, w, i	Indices for Season, Week, and Contestant
$J_{s,w,i}$	Total judges' score for contestant i in season s, week w
$p_{s,w,i}$	Estimated **Fan Vote Share** ($\sum_i p_{s,w,i} = 1$)
$u_{s,w,i}$	Latent utility (popularity strength) of contestant
$\mathcal{E}_{s,w}$	Set of eliminated contestants in week w
$\mathcal{K}_{s,w}$	Set of safe (kept) contestants in week w
T	Softmax temperature parameter
CR	Consistency Ratio in AHP
ϕ_j	SHAP value for feature j

3 Data Description and Processing

3.1 Data Overview

The dataset provided (**2026_MCM_Problem_C_Data.csv**) contains comprehensive records for Seasons 1 through 34. Key attributes include contestant informtion (Name, Partner, Age, Industry), weekly judge scores, and placement outcomes.

3.2 Feature Engineering and Preprocessing

We constructed a robust data pipeline to transform raw logs into analytical tensors:

- **Handling Missing Values:** Entries marked as 'N/A' are excluded. Eliminated contestants are tracked via the `exit_week` and `exit_type` derived fields.
- **Normalization:** Judge scores are normalized to a 0-1 scale to allow comparison across seasons (some seasons used max 30, others 40).
- **Feature Extraction:** We extracted static features (Age, Gender, Industry Category) and dynamic features (Running Average Score, Momentum).

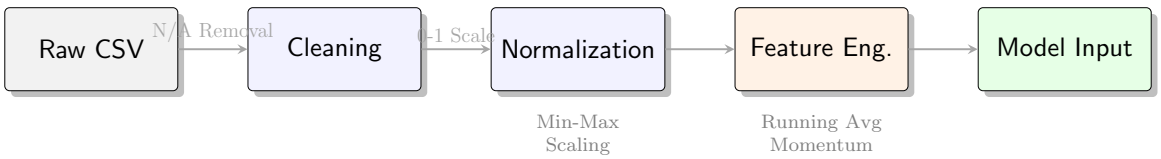


Figure 2: **Data Processing Pipeline.**

4 Task 1: Bayesian Estimation of Fan Vote Percent

We reconstruct the *hidden* weekly fan voting signal as a Bayesian inverse problem. Let $\mathbf{p}_{s,w} = \{p_{s,w,i}\}_{i \in \mathcal{A}_{s,w}}$ be the (unknown) **fan vote percent** over the active set $\mathcal{A}_{s,w}$ at week w in season s. Given judges' scores J and observed eliminations E, we infer

$$P(\mathbf{p} \mid \mathcal{D}) \propto P(\mathcal{D} \mid \mathbf{p})\, P(\mathbf{p}), \tag{1}$$

where the likelihood encodes the elimination rule and the prior enforces temporal smoothness.

4.1 Problem 1: Model Formulation

Latent popularity dynamics. We introduce latent utilities $u_{s,w,i}$ (log-popularity) and use a random-walk prior:

$$u_{s,w,i} \sim \mathcal{N}(u_{s,w-1,i}, \sigma^2_{\text{RW}}), \tag{2}$$
$$u_{s,w,i} \sim \mathcal{N}(0, \sigma^2_{\text{level}}), \tag{3}$$
$$u_{s,1,i} \sim \mathcal{N}(0, \sigma^2_{u0}). \tag{4}$$

Utilities map to vote percent via softmax (temperature $T = 0.25$):

$$p_{s,w,i} = \frac{\exp(u_{s,w,i}/T)}{\sum_{j \in \mathcal{A}_{s,w}} \exp(u_{s,w,j}/T)}. \tag{5}$$

Elimination-rule likelihood. Let $\sigma(x) = \frac{1}{1+\exp(-x)}$. For each week, define eliminated set $\mathcal{E}_{s,w}$ and kept set $\mathcal{K}_{s,w}$ (withdrawals excluded).

Rank era. Let $r^{(J)}_{s,w,i}$ be the rank of judges' totals (1=best) and $r^{(F)}_{s,w,i}$ be the rank implied by $p_{s,w,i}$. The combined rank is

$$b_{s,w,i} = r^{(J)}_{s,w,i} + r^{(F)}_{s,w,i} + 0.01\, r^{(F)}_{s,w,i}. \tag{6}$$

For $e \in \mathcal{E}_{s,w}$ and $k \in \mathcal{K}_{s,w}$,

$$P(e \prec k \mid \mathbf{p}, J) = \sigma\left(\frac{b_{s,w,e} - b_{s,w,k}}{\tau_{\text{rank}}}\right), \tag{7}$$

and the weekly likelihood is

$$P(\mathcal{E}_{s,w} \mid \mathbf{p}, J) \propto \prod_{e \in \mathcal{E}_{s,w}} \prod_{k \in \mathcal{K}_{s,w}} \sigma\left(\frac{b_{s,w,e} - b_{s,w,k}}{\tau_{\text{rank}}}\right). \tag{8}$$

Percent era. Let $q_{s,w,i}$ be the normalized judge percent and define

$$c_{s,w,i} = q_{s,w,i} + p_{s,w,i}. \tag{9}$$

Then

$$P(e \prec k \mid \mathbf{p}, J) = \sigma\left(\frac{c_{s,w,k} - c_{s,w,e}}{\tau_{\text{pct}}}\right), \tag{10}$$

$$P(\mathcal{E}_{s,w} \mid \mathbf{p}, J) \propto \prod_{e \in \mathcal{E}_{s,w}} \prod_{k \in \mathcal{K}_{s,w}} \sigma\left(\frac{c_{s,w,k} - c_{s,w,e}}{\tau_{\text{pct}}}\right). \tag{11}$$

Posterior samples of $\mathbf{p}$ are obtained by Metropolis–Hastings MCMC over $\{u_{s,w,i}\}$.

4.2 Problem 2: Consistency Evaluation

We measure how often posterior draws reproduce the observed elimination under the rule. For Percent-style rules, the draw-specific margin is

$$\Delta^{(m)}_{s,w} = \min_{k \in \mathcal{K}_{s,w}} \left(q_{s,w,k} + p^{(m)}_{s,w,k}\right) - \max_{e \in \mathcal{E}_{s,w}} \left(q_{s,w,e} + p^{(m)}_{s,w,e}\right). \tag{12}$$

The weekly consistency score is the posterior predictive probability:

$$\text{Consistency}_{s,w} = P(\Delta_{s,w} \geq 0 \mid \mathcal{D}) \approx \frac{1}{M} \sum_{m=1}^{M} \mathbb{I}\left[\Delta^{(m)}_{s,w} \geq 0\right]. \tag{13}$$

Rank vs. Percent consistency. Across all seasons, Percent rules yield higher consistency than Rank rules, while Rank+Save is intermediate (Figure 3). The heatmap in Figure 4 localizes low-consistency weeks, concentrated in Rank-based eras.

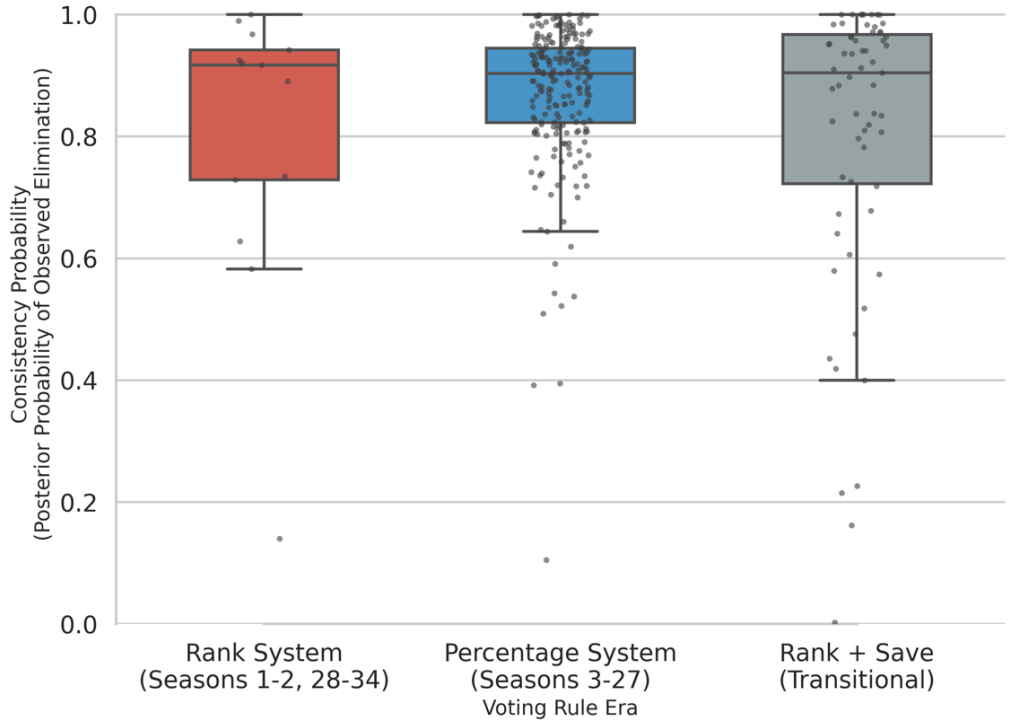


Figure 3: **Structural Integrity Analysis.** Weekly consistency distributions under Rank, Percentage, and Rank+Save.

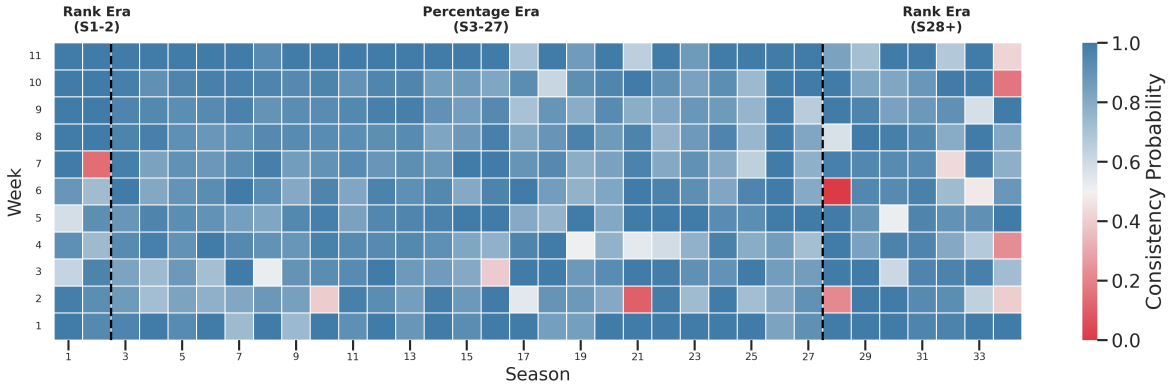


Figure 4: **The Heatmap of weekly consistency across seasons and voting rules.**

4.3 Problem 3: Uncertainty Assessment

We quantify posterior uncertainty using the relative 90% credible interval width (RCI90). Let

$$p_{s,w,i}^{(0.05)} = \text{Quantile}_{0.05}(p_{s,w,i} \mid \mathcal{D}), \quad p_{s,w,i}^{(0.95)} = \text{Quantile}_{0.95}(p_{s,w,i} \mid \mathcal{D}), \tag{14}$$

$$\text{CI90}_{s,w,i} = p_{s,w,i}^{(0.95)} - p_{s,w,i}^{(0.05)}, \tag{15}$$

$$d_{s,w,i} = \max\left(\mathbb{E}[p_{s,w,i} \mid \mathcal{D}], \frac{1}{|\mathcal{A}_{s,w}|}\right), \tag{16}$$

$$\text{RCI90}_{s,w,i} = \frac{\text{CI90}_{s,w,i}}{d_{s,w,i}}. \tag{17}$$

Temporal trend. Week-level uncertainty decreases as eliminations accumulate:

$$\text{CI90}_{s,w} = \frac{1}{|\mathcal{A}_{s,w}|} \sum_{i \in \mathcal{A}_{s,w}} \left(p_{s,w,i}^{(0.95)} - p_{s,w,i}^{(0.05)}\right). \tag{18}$$

See Figure 5.

Uncertainty vs. support. Across contestant-week pairs, higher support tends to have smaller relative uncertainty:

$$\rho = \text{Corr}(\mathbb{E}[p_{s,w,i} \mid \mathcal{D}], \text{RCI90}_{s,w,i}), \tag{19}$$

illustrated in Figure 6.

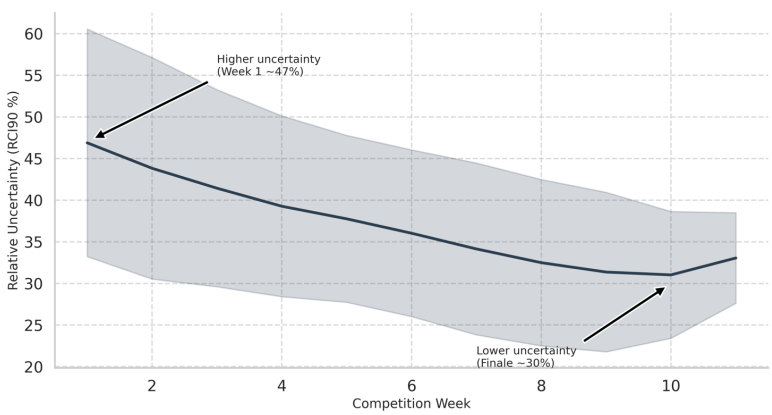


Figure 5: **Temporal Convergence.** Week-level uncertainty decreases over the season.

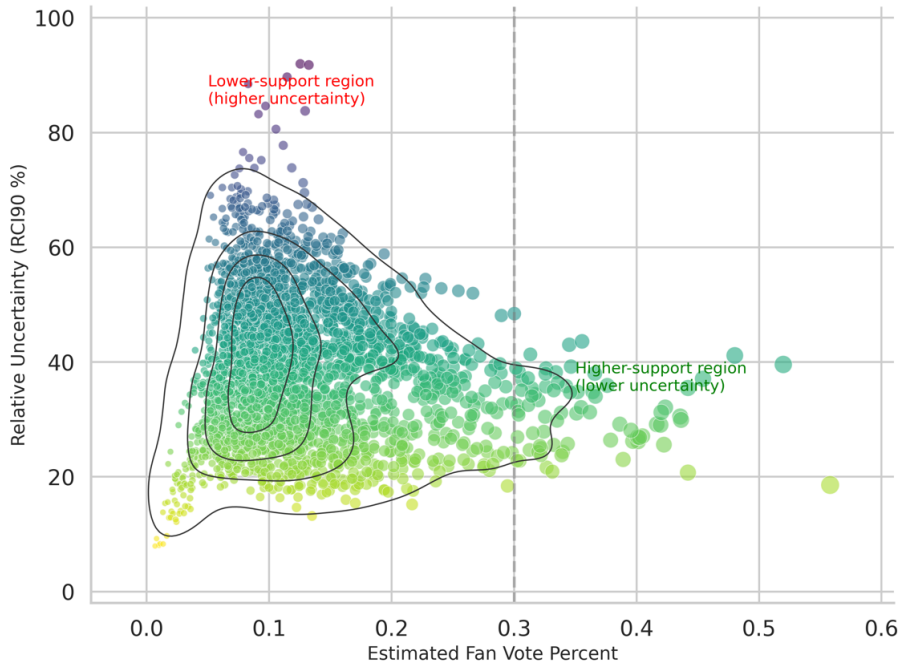


Figure 6: **The Anatomy of Risk.** Relative uncertainty vs. fan vote percent.

4.4 Evidence: Case Studies

Because the active pool shrinks over time, the uniform baseline rises:

$$p_{s,w}^{\text{uni}} = \frac{1}{N_{s,w}}, \quad N_{s,w} = |\mathcal{A}_{s,w}|. \tag{20}$$

We therefore compare *relative* fan support against judge percent via

$$\Delta_{s,w,i}^{(F-J)} = \mathbb{E}[p_{s,w,i} \mid \mathcal{D}] - q_{s,w,i}. \tag{21}$$

Jerry Rice (Season 2): "Gap Expansion". Rice's fan support increasingly exceeds technical merit over weeks (Figure 7).

Bobby Bones (Season 27): "Relative Collapse". With a shrinking field, near-constant judge percent implies declining relative technical standing:

$$q_{s,w,\text{Bobby}} < p_{s,w}^{\text{uni}} = \frac{1}{N_{s,w}}, \tag{22}$$

while fan support remains strong (Figure 8).

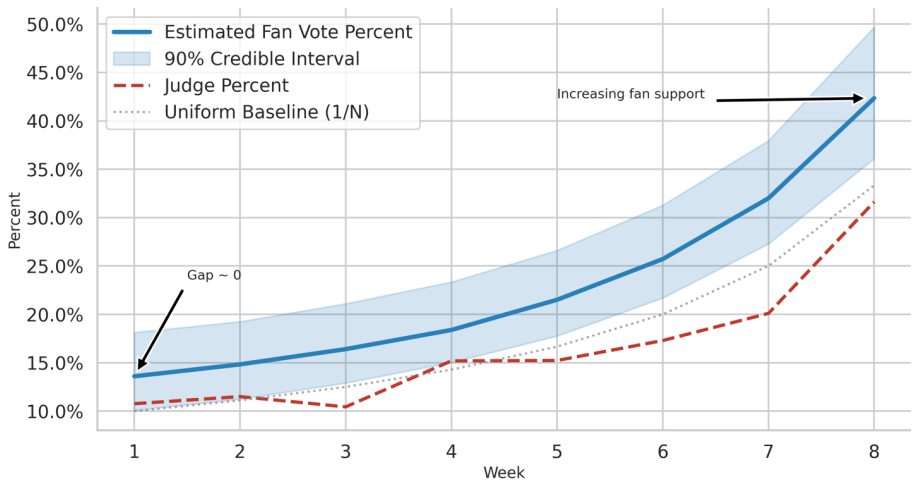


Figure 7: **The "Teflon Star" Trajectory.** Jerry Rice: judge percent and inferred fan vote percent over time.

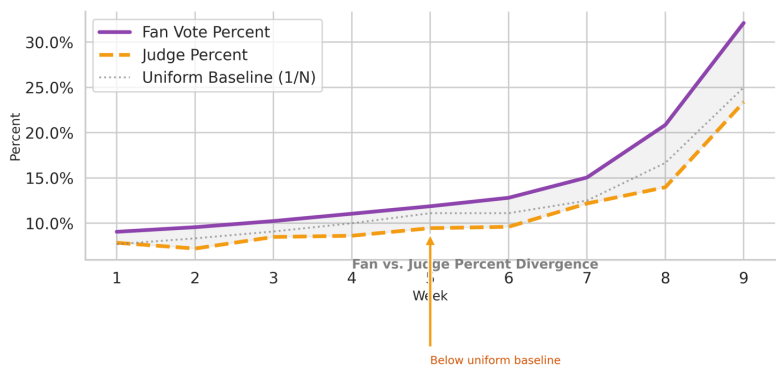


Figure 8: **The Relative Collapse.** Bobby Bones: judge percent and inferred fan vote percent over time.

5 Task 2: Voting Schemes Comparison

5.1 Comparative Analysis: Rank v.s. Percent

Using the inferred fan-vote percent from Task 1, we conduct a weekly counterfactual reconstruction across all seasons. Keeping judges' scores fixed, we recompute weekly rankings under the Rank and Percent aggregation rules and compare them using Kendall's τ and Spearman's ρ.

Ranking agreement metrics. Kendall's rank correlation is defined as

$$\tau = \frac{n_c - n_d}{\frac{1}{2}n(n-1)},$$

and Spearman's rank correlation as

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)},$$

where d_i denotes the rank difference of contestant i under the two rules.

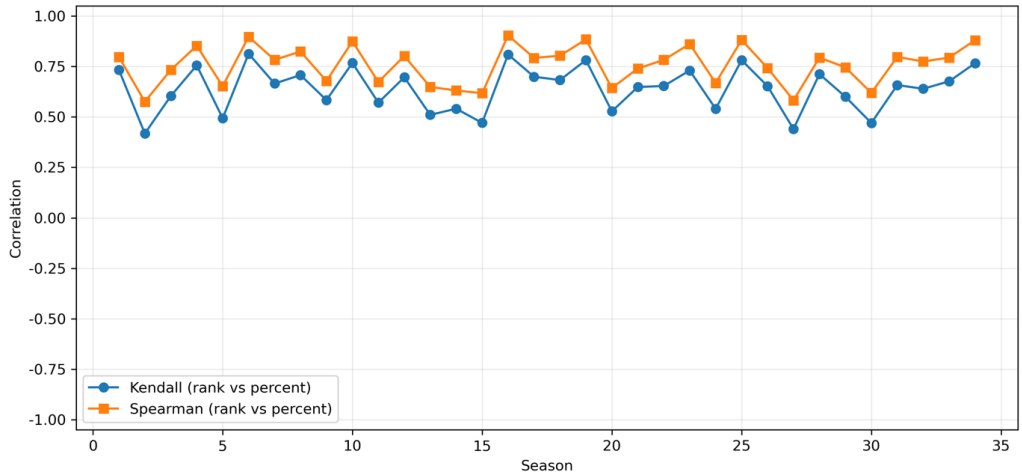


Figure 9: Season-level Kendall and Spearman Coefficient Between Rank and Percent Voting Schemes

As shown in Figure 9, the two mechanisms exhibit strong overall agreement: the mean Kendall coefficient is **0.692** and the mean Spearman coefficient is **0.783**, indicating that most macro-level ranking structures are preserved.

Top-k stability. To assess robustness among high-ranked contestants, we define the Top-k overlap ratio

$$\text{Top-}k = \frac{|Top_k(A) \cap Top_k(B)|}{|Top_k(A) \cup Top_k(B)|}.$$

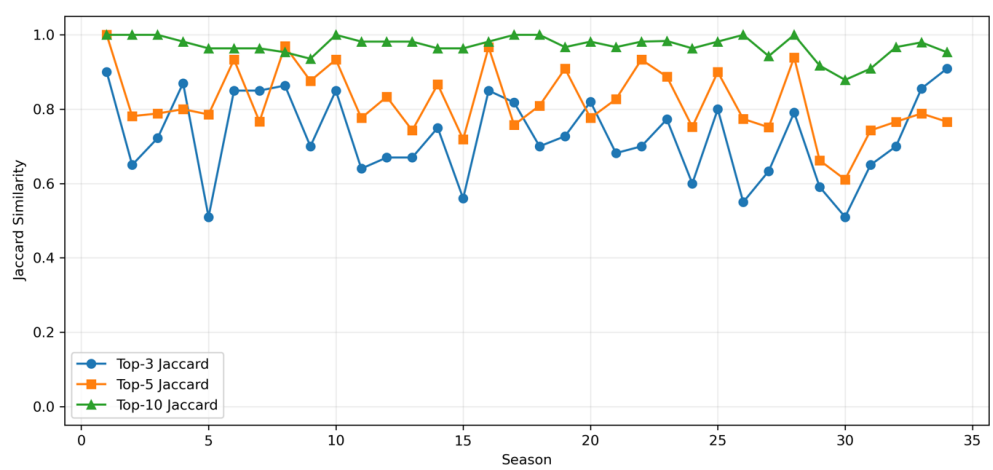


Figure 10: Season-level Top-k Ratio Between Rank and Percent Voting Schemes

Figure 10 shows that overlap increases monotonically with k, from **0.735** (Top-3) to **0.844** (Top-5), and reaches **0.982** at Top-10, suggesting that rule choice mainly affects fine-grained ordering at the very top.

5.2 Fan Favor Analysis

We further compare the two schemes along **fan alignment** and **sensitivity**.

Fan alignment. Let v_w denote fan vote shares in week w, and R_w the resulting ranking. Fan alignment is measured as

$$A_w^{rank} = \rho(R_w^{rank}, -v_w), \quad A_w^{pct} = \rho(R_w^{pct}, -v_w).$$

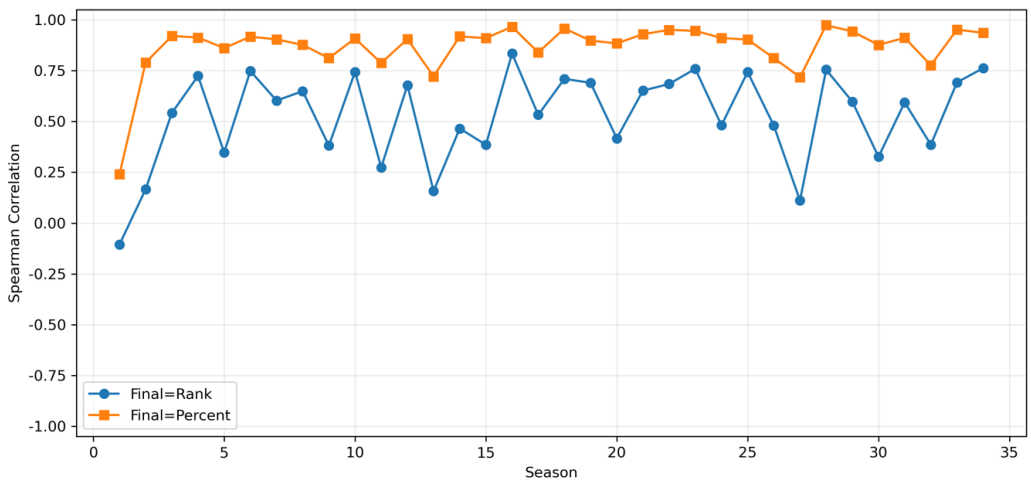


Figure 11: Season-level Fan Alignment Between Rank and Percent Voting Schemes

As shown in Figure 11, the Percent rule exhibits substantially higher fan alignment (mean **0.740**) than the Rank rule (mean **0.328**), indicating that Percent aggregation transmits fan preferences more directly.

Sensitivity analysis. To evaluate robustness under fan-vote uncertainty, we perform Monte Carlo perturbations:

$$v_{w,i}^{(b)} \sim \text{TruncNormal}(\mu_{w,i}, \sigma_{w,i}; 0, 1),$$

and define ranking sensitivity as

$$S_w = \mathbb{E}_b\big[1 - \tau(R_w, R_w^{(b)})\big] .$$

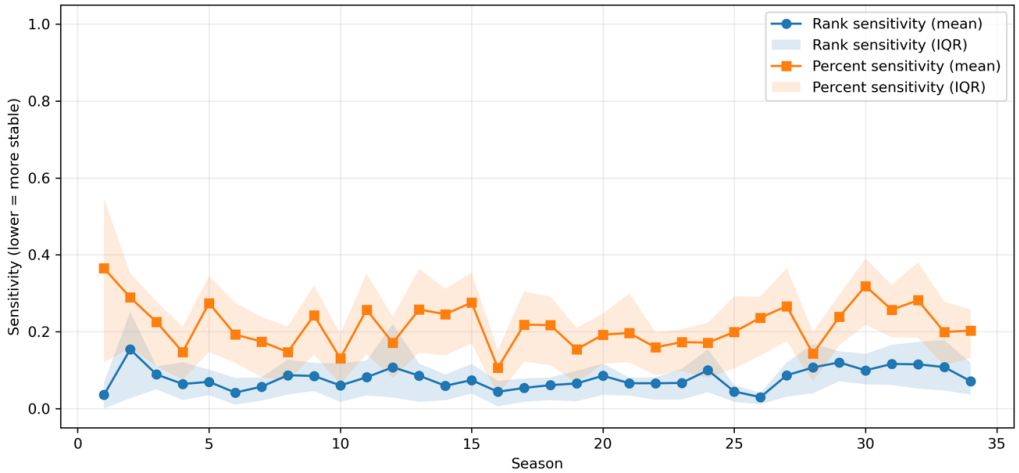


Figure 12: Season-level Sensitivity Analysis Between Rank and Percent Voting Schemes

Figure 12 shows that Percent is more sensitive to vote perturbations (mean **0.218**) than Rank (mean **0.112**), reflecting the stabilizing effect of ordinal aggregation.

5.3 Case Studies of Four Controversial Contestants

To quantify systematic divergence between fans and judges, we define the weekly score gap

$$D_{w,i}^{score} = v_{w,i} - j_{w,i},$$

and the season-average support gap

$$\bar{D}_i = \frac{1}{T_i} \sum_{w=1}^{T_i} (v_{w,i} - j_{w,i}).$$

We analyze Jerry Rice, Billy Ray Cyrus, Bristol Palin, and Bobby Bones. All four exhibit significantly positive support gaps (**0.11–0.18**), indicating persistent fan-driven advantages despite weaker judges' evaluations.

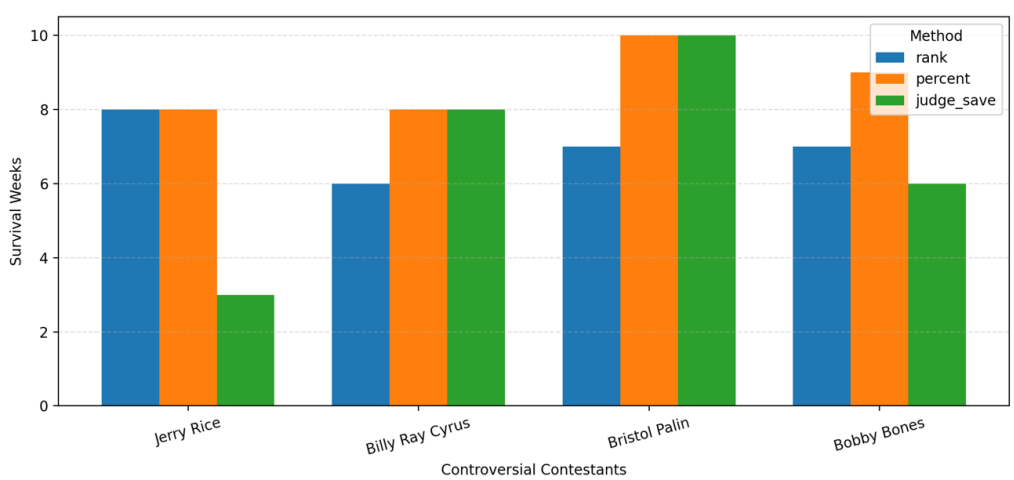


Figure 13: Case Study of Four Controversial Contestants

Figure 13 further shows that Percent voting yields the longest survival durations, whereas Rank shortens advancement and Judge-save disproportionately penalizes fan-driven contestants.

5.4 AHP: Evaluation of the Three Voting Schemes

We incorporate Rank, Percent, and Judge-save into a multi-criteria evaluation using the **Analytic Hierarchy Process (AHP)**. The Judge-save rule mimics the Bottom-2 protection mechanism:

$$\text{Bottom2} = \arg\min_i (T_i^{percent}), \quad \text{Eliminate} = \arg\min_i (J_i^{percent}).$$

Although Judge-save improves expert control, its elimination hit rate (**0.524**) is lower than Percent (**0.904**) and Rank (**0.719**), indicating a trade-off between consistency and intervention.

We evaluate methods along four criteria: **Fan Alignment, Stability, Elimination Accuracy, and Transparency**. The pairwise comparison matrix A satisfies

$$Aw = \lambda_{\max} w, \quad \sum_k w_k = 1,$$

with consistency ratio $CR = 0.0115 < 0.1$, confirming reliable weighting.

Table 1: AHP Criteria Weights

Criterion	Fan Alignment	Stability	Elimination Accuracy	Transparency
Weight	0.2772	0.1601	0.4673	0.0954

Table 2: Overall Scores of Voting Methods

Method	Percent	Rank	Judge-save
Score	0.8399	0.4955	0.2772

Based on the AHP results, the **Percent** scheme achieves the highest overall score, followed by **Rank**, while **Judge-save** ranks last.

6 Task 3: Impact Analysis of Celebrity Characteristics Based on Random Forest

6.1 Research Objectives and Problem Statement

Task 3 quantifies how **ballroom partners** and **celebrity attributes** (age, industry, region) affect competition outcomes, and compares whether these effects differ for **judges' scoring** vs. **fan voting**. Given the nonlinear and mixed-type structure of the data, we use a **Random Forest** regressor and interpret feature contributions via **SHAP**.

6.2 Data Structure and Feature Engineering

The raw data are at the contestant-season level; we convert them to a **week-level panel** aligned with voting/elimination. Key constructed variables include *judge_sum*, *judge_pct*, and *active_count* to normalize across seasons with different scales. To capture inertia while avoiding leakage, we use only lagged information:

$$prev_judge_pct_{i,t} = judge_pct_{i,t-1}$$

$$prev_fan_share_{i,t} = fan_share_{i,t-1}$$

$$cummean_fan_share_{i,t} = \frac{1}{t-1}\sum_{k=1}^{t-1} fan_share_{i,k}$$

We train two separate models to isolate mechanism differences:

Fans model (predicting fan vote percent). Numerical: `age, season, week, active_count, judge_pct, prev_judge_pct, prev_fan_share, cummean_fan_share`.
Categorical: `industry, homecountry/region, ballroom_partner`.

Judges model (predicting judge score percent). Numerical: `age, season, week, active_count, prev_judge_pct`.
Categorical: `industry, homecountry/region, ballroom_partner`.
(Fan-history variables are excluded to avoid causal confounding.)

6.3 Random Forest Regression Model

Random Forest aggregates B bootstrap trees:

$$\hat{y} = \frac{1}{B}\sum_{b=1}^{B} T_b(x). \tag{23}$$

We apply a unified preprocessing pipeline (median imputation for numerical features, mode imputation + one-hot encoding for categorical features), and adopt **Group K-Fold** by season to prevent cross-season leakage.

Performance is evaluated by

$$R^2 = 1 - \frac{\sum(y-\hat{y})^2}{\sum(y-\bar{y})^2}, \tag{24}$$

and

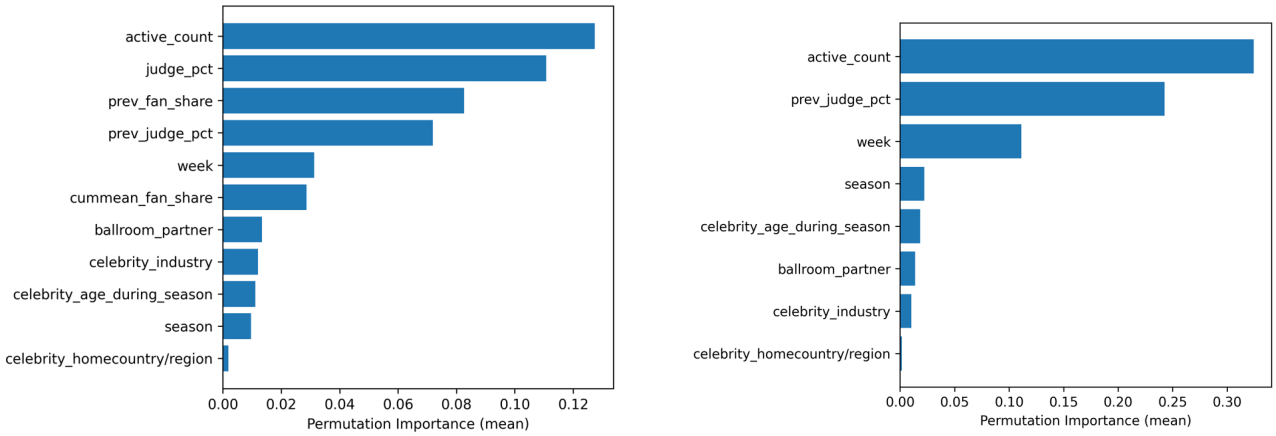
$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|. \tag{25}$$

Table 3 shows that the Judges model is more regular (higher R^2 and lower MAE), while fan voting is noisier and more volatile.

Table 3: The Performance of Random Forest Models

Random Forest Model	R^2	MAE
Fan	0.8122	0.0151
Judge	0.8907	0.0118

Permutation importance. Figure 14 reports permutation importance for both models. Structural/history features dominate predictive power, but *ballroom_partner* and celebrity attributes (age/industry/region) still contribute consistently, indicating measurable (though secondary) effects on outcomes.



(a) Fan Vote Model (b) Judge Score Model

Figure 14: Permutation Importance Analysis for Random Forest Model

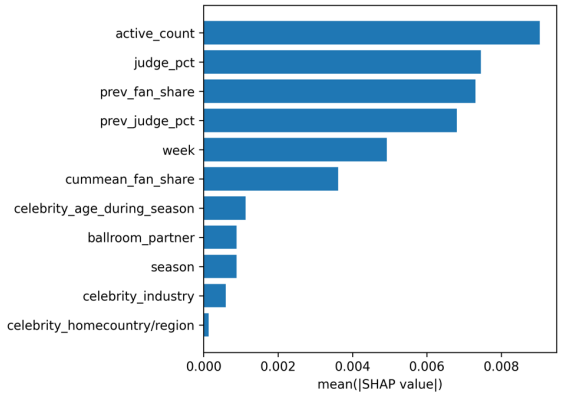
6.4 SHAP Importance Analysis

To interpret marginal effects, SHAP decomposes predictions as

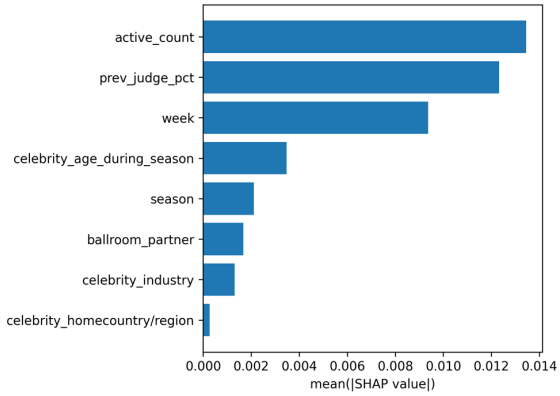
$$f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i, \tag{26}$$

where ϕ_i is the contribution of feature i.

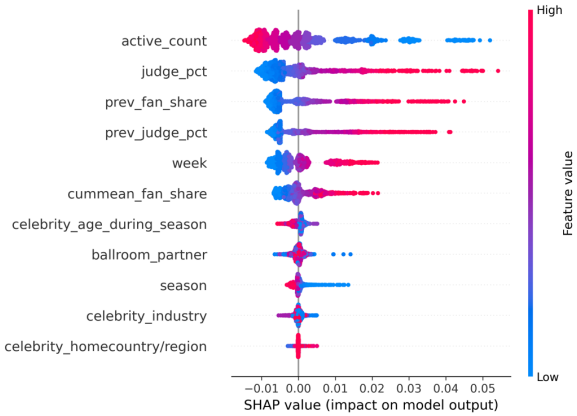
Figure 15 summarizes SHAP patterns: (i) *ballroom_partner* exhibits a dispersed contribution distribution, implying heterogeneous partner effects; (ii) *celebrity_age_during_season* shows a moderate directional influence; (iii) *celebrity_industry* is more pronounced in the Fans model (public familiarity), while *homecountry/region* remains smaller but stable. Overall, partner assignment and celebrity background provide supplementary explanatory power beyond structural competition dynamics.



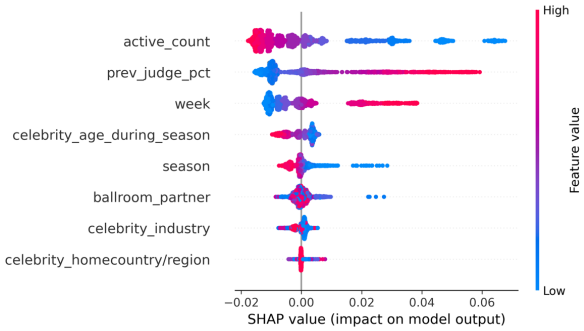
(a) SHAP Feature Importance for Fan Vote Model



(b) SHAP Feature Importance for Judge Score Model



(c) SHAP Beeswarm for Fan Vote Model



(d) SHAP Beeswarm for Judge Score Model

Figure 15: SHAP-Based Feature Contribution Analysis for Random Forest Models

7 Task 4: DW-UW System Design and Evaluation

7.1 Motivation: Fairness–Engagement Trade-off

The core challenge of *Dancing with the Stars* is balancing **technical fairness** (judges' evaluations) with **audience engagement** (fan vote percent). Existing Rank- or Percent-based systems rely on fixed weights and therefore fail to adapt to the *quality* of the fan signal: noisy fan votes may dominate outcomes, while stable fan preferences may be underweighted.

Our solution is the **DWUW system**, which dynamically adjusts judge influence according to the inferred uncertainty of fan voting.

7.2 System Design

Adaptive Weighting. Instead of a fixed split, the weekly judge weight w_t is defined as a monotone function of fan vote uncertainty U_t:

$$w_t = w_{\min} + (w_{\max} - w_{\min}) \cdot \frac{U_t}{U_t + \kappa}, \tag{27}$$

with $w_{\min} = 0.35$, $w_{\max} = 0.70$, and $\kappa = 0.02$. Higher uncertainty increases judge influence, while stable fan signals grant greater audience power. Figure 16 illustrates the mapping and empirical distribution of U_t.

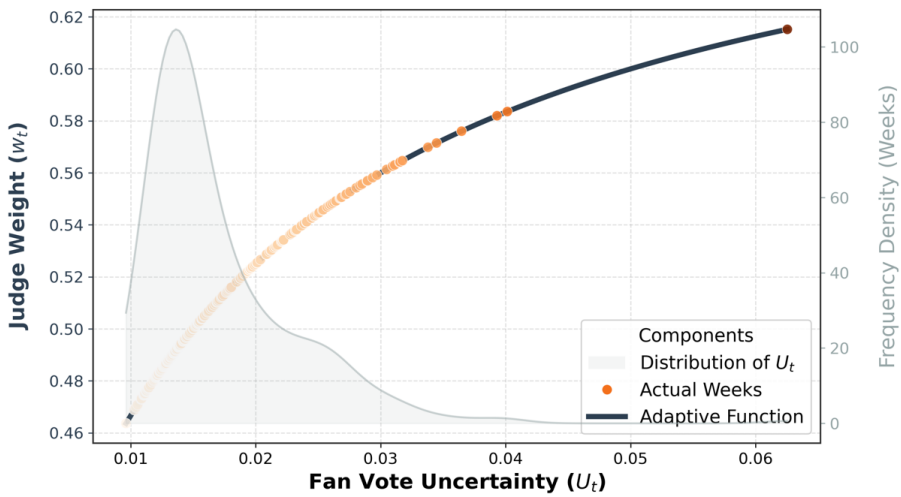


Figure 16: **Adaptive Weighting.** Judge weight w_t as a function of uncertainty U_t.

Conditional Judges Save. To prevent eliminations that sharply contradict technical merit, we introduce a conditional judges-save mechanism: when the Bottom-2 contestants exhibit a significant technical gap ($q_{\text{safe}} - q_{\text{elim}} > \delta = 0.02$), judges retain the stronger dancer. This intervention is rare (about $\sim$15% of weeks) and preserves suspense.

7.3 Evaluation Results

We evaluated DWUW across Seasons 1–34 and compared it with historical rules.

Fairness. DWUW distinguishes outcomes driven by noisy fan signals from those supported by stable audience preference. Figure 17 summarizes representative corrections and validations: fan-driven anomalies (e.g., Bobby Bones, Season 27) are moderated under high uncertainty, while technically superior contestants are preserved when fan signals are reliable.

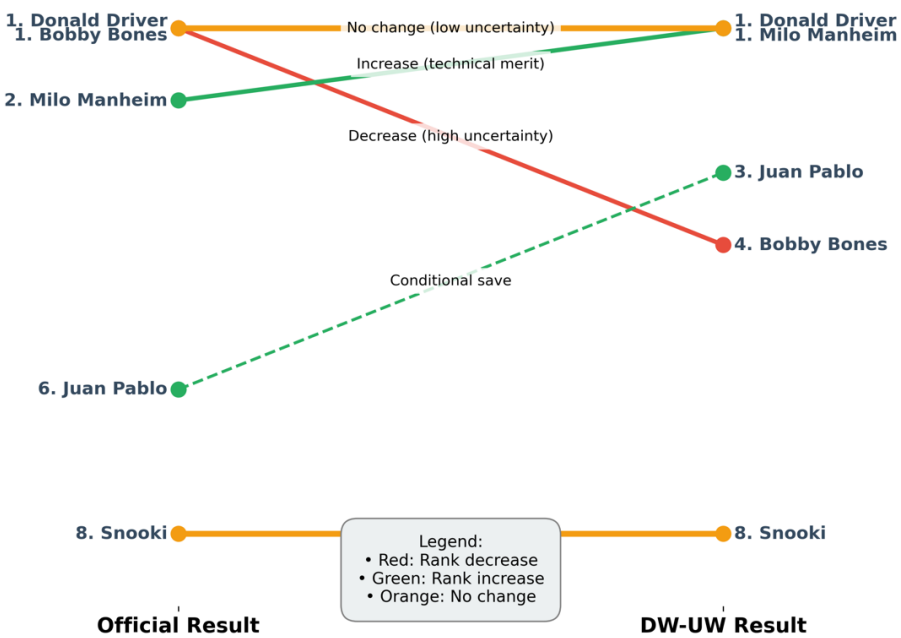


Figure 17: **Impact Spectrum.** Selected season outcomes under DWUW relative to the original results.

Engagement. Audience participation is quantified by the **upset rate**, while fairness is measured via the Spearman correlation between final placement and technical skill. As shown in Figure 18, DWUW achieves strong fairness (mean $|r_s| \approx 0.98$) while maintaining a high upset rate (about ∼68%), indicating sustained fan influence rather than deterministic judge dominance.

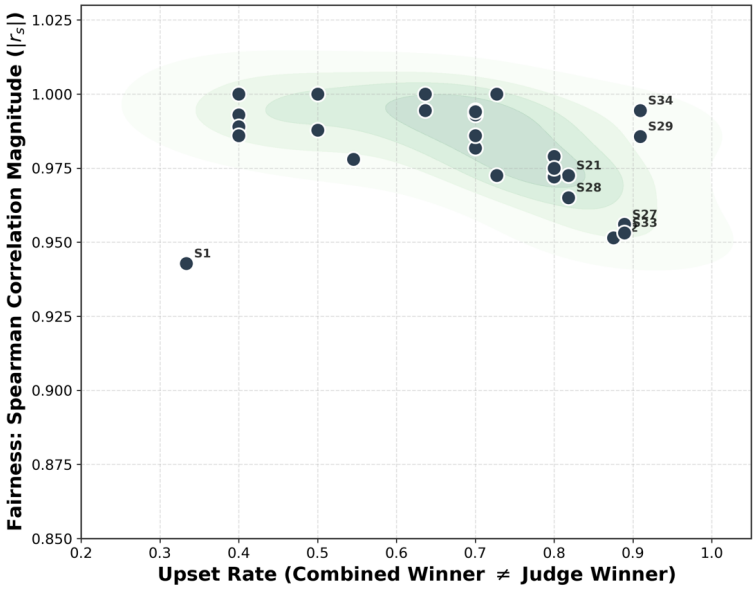


Figure 18: **Fairness vs. Upset Rate.** Season-level results under historical rules and DWUW.

Suspense. We further examined the score gap between the lowest safe and eliminated contestants. Figure 19 shows that the median safety margin remains small (median $\approx$ 0.013), demonstrating that DWUW does not artificially separate outcomes near the elimination boundary.

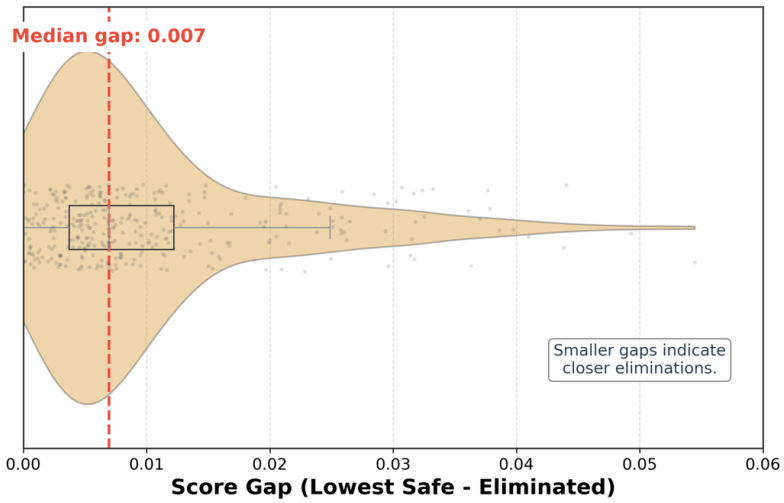


Figure 19: **Suspense Distribution.** Distribution of weekly safety margins under DWUW.

7.4 Recommendation

We recommend adopting the **DWUW system** for future seasons. It adaptively increases judge influence when fan vote uncertainty is high, preserves audience impact when signals are stable, and can be implemented using only judge scores, fan vote percent, and posterior uncertainty estimates. The adaptive rule can be communicated transparently as a response to signal quality rather than subjective intervention.

8 Computational Implementation and Code Availability

All computational analyses and simulations in this paper were implemented in Python and are fully reproducible. The complete source code, including data preprocessing, Bayesian inference (MCMC), voting-rule simulations, Random Forest modeling, SHAP analysis, and figure generation, is publicly available at:

`https://github.com/julian-wu-cs/MCM2026-C-2619940`

Memo to DWTS Producers

To: Executive Producers, *Dancing with the Stars*
From: Team #2619940
Date: February 2026
Subject: Improving the Combination of Judge Scores and Fan Votes

Audience. This memo is written for show producers and standards teams. It is intentionally non-technical: it translates modeling results into actionable rule choices, communication guidance, and guardrails.
What we learned. The way you combine judges' scores and fan votes directly changes eliminations. A **rank-based** rule is simple and stable but can create "step effects" late in the season (small changes flip whole ranks). A **percent-based** rule better reflects true margins (how big the lead is), but it can be vulnerable to noisy voting weeks (low turnout, tight margins, and campaign shocks).
Producer implications.

- A single fixed split treats a razor-thin fan margin the same as a landslide, which can inflate early-week randomness and drive "unfair" eliminations.
- Late-season episodes are most sensitive because the contestant pool is small; rule "step effects" become largest exactly when stakes are highest.
- Viewer trust depends less on the exact split and more on whether the rule is **consistent, explainable, and visibly prevents obvious technical mismatches**.

Recommendations.

1. **Use Percent shares as the default.** Combine "share of judges' points" with "share of fan votes" so margins matter (not just order).
2. **Make the split adaptive using a published "confidence" indicator.** When voting is noisy (low turnout and/or tiny Bottom-2 vote margin), temporarily lean more on judges; when voting is clear, lean more on fans. This can be computed internally from vote totals without releasing any counts.
3. **Keep a narrowly defined Judges' Save as a safety net.** Use it only when the Bottom 2 have a clear technical gap, so it corrects obvious mismatches but does not override close contests.

How to communicate it on-air. Announce that "the show adjusts how much weight judges carry based on voting confidence" and display a simple confidence meter (High/Medium/Low). This frames adaptation as transparency and fairness rather than producer intervention.
Bottom line. This approach protects strong dancers from noisy early eliminations, preserves fan influence when it is clear, and gives producers a transparent rule they can defend on-air.

Yours Sincerely,
Team 2619940

References

References

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[2] Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30. (The foundational paper for SHAP values used to analyze celebrity characteristics and partner impact.)

[3] Caron, F., & Teh, Y. W. (2012). Bayesian nonparametric models for rank data. *Bayesian Analysis*, 7(3), 613–666. (Relevant to the "Inverse Problem" of reconstructing latent preferences from observed rankings.)

[4] Shoham, Y., & Leyton-Brown, K. (2008). *Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations*. Cambridge University Press. (Supports the adaptive mechanism design and weight optimization in the proposed DW-UW system.)

[5] Plassmann, F., & Tideman, N. (2011). Determinants of success in "Dancing with the Stars": The role of judges' scores and fan votes. *Journal of Quantitative Analysis in Sports*. (Directly addresses the dual-voting system controversy and the balance between technical skill and popularity.)

[6] West, M., Harrison, P. J., & Migon, H. S. (1985). Dynamic generalized linear models and Bayesian forecasting. *Journal of the American Statistical Association*, 80(389), 73–83. (Relates to the Dynamic Bayesian Network and random-walk prior used to model fan base stability.)

Report on Use of AI

1. OpenAI ChatGPT (Jan 30, 2026, version ChatGPT-5.2)

Query1: Please explain the difference between the Random Forest Model and the XGBoost.

Output: Random Forest and XGBoost are both ensemble learning methods based on decision trees, but they differ in their training strategies and optimization mechanisms. Random Forest is a bagging-based method that builds multiple independent decision trees using bootstrap sampling and random feature selection, and then aggregates their predictions by averaging (for regression) or voting (for classification). Its main goal is to reduce variance and improve model stability. In contrast, XGBoost (Extreme Gradient Boosting) is a boosting-based method that builds trees sequentially, where each new tree is trained to correct the residual errors of the previous ones. XGBoost incorporates gradient-based optimization, regularization terms, and advanced techniques such as learning rate shrinkage and tree pruning, which help reduce bias and overfitting while improving predictive performance. Overall, Random Forest emphasizes parallelism and robustness, whereas XGBoost focuses on sequential optimization and higher predictive accuracy.

2. OpenAI ChatGPT (Jan 30, 2026, version ChatGPT-5.2)

Query2: Please tell the pros and cons of the AHP Model

Output: The Analytic Hierarchy Process (AHP) is a multi-criteria decision-making method that decomposes complex problems into a hierarchical structure of goals, criteria, and alternatives. One major advantage of AHP is its ability to incorporate both qualitative and quantitative criteria, making it suitable for real-world decision problems with subjective judgments. It also provides a consistency check mechanism to evaluate the reliability of pairwise comparisons, which enhances decision transparency and interpretability. However, AHP also has limitations. It relies heavily on expert judgments, which may introduce subjectivity and bias.